Osaid Khan

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Education

Pace University — New York City, NY
Master of Science (MS) in Computer Science (GPA - 4.0/4.0) — *Jan 2024 - Dec 2025*

Dr. A.P.J. Abdul Kalam Technical University — Noida, India
Bachelor of Technology (B.Tech) in Computer Science — *Jul 2016 - Sep 2020*

Availability: Available to onboard by December 2026; graduation date: December 2025

Technical Skills

Languages & GPU: C/C++, Python, CUDA programming, GPU hardware architecture, GPU memory models, Stream asynchronous scheduling, GPU/NPU hardware adaptability
Inference Optimization: Deep learning inference compilation, operator fusion, computational graph optimization, constant folding, memory reuse, cache optimization, scheduling optimization, quantization compilation
Parallelism: Tensor parallelism (TP), pipeline parallelism (PP), sequence parallelism, MoE expert parallelism, multi-card communication mechanisms
Operators & Profiling: Matrix operations, normalization, activation functions, memory access optimization, vectorization acceleration, precision alignment, software-hardware collaborative optimization, NVIDIA Nsight, Profiler
Frameworks & Systems: vLLM, SGLang, TensorRT-LLM, PyTorch, LangGraph, FastAPI, OpenTelemetry, Docker, Kubernetes, AWS

Professional Experience

Software Engineer — Phoenix, AZ
Sperse — *Feb 2026 - Present*

- Optimized end-to-end inference latency for a **Python/FastAPI** multi-agent platform by profiling OpenTelemetry traces across model and tool pipelines, eliminating scheduling bottlenecks while serving 10,000+ users through 40+ specialized agents
- Designed distributed parallel execution for long-running AI workloads with **Temporal**, Redis, and Kubernetes; collaborated cross-team to document and present Profiler results, own complex issue resolution under pressure, and improve throughput and high-concurrency stability

Software Engineer — Noida, India
Qualitest — *Feb 2021 - Dec 2023*

- Improved memory access and query throughput in a Java/Spring Boot storage engine by optimizing **PostgreSQL** window functions, materialized views, and indexes, then surfaced performance data through React visualizations
- Orchestrated 100K+ weekly parallel executions with **AWS Step Functions** and Lambda, removing timeout bottlenecks and improving scheduling efficiency, load balance, and failure recovery under production concurrency

Projects

Agentic CUDA Optimization Virtual Lab | *CUDA, C++, Python, LangGraph, NVIDIA Nsight* — github.com/ozzyozbourne

- Built an agentic optimization loop for handwritten **CUDA** kernels (matrix operations, normalization, and activation functions), combining evolutionary search with **NVIDIA Nsight** profiling to identify memory access, computing pipeline, and scheduling bottlenecks
- Implemented operator fusion, vectorization acceleration, and precision alignment experiments; benchmarked single-card inference throughput, GPU memory usage, and inference latency against **vLLM** and **TensorRT-LLM** while evaluating TP, PP, sequence, and MoE parallel strategies